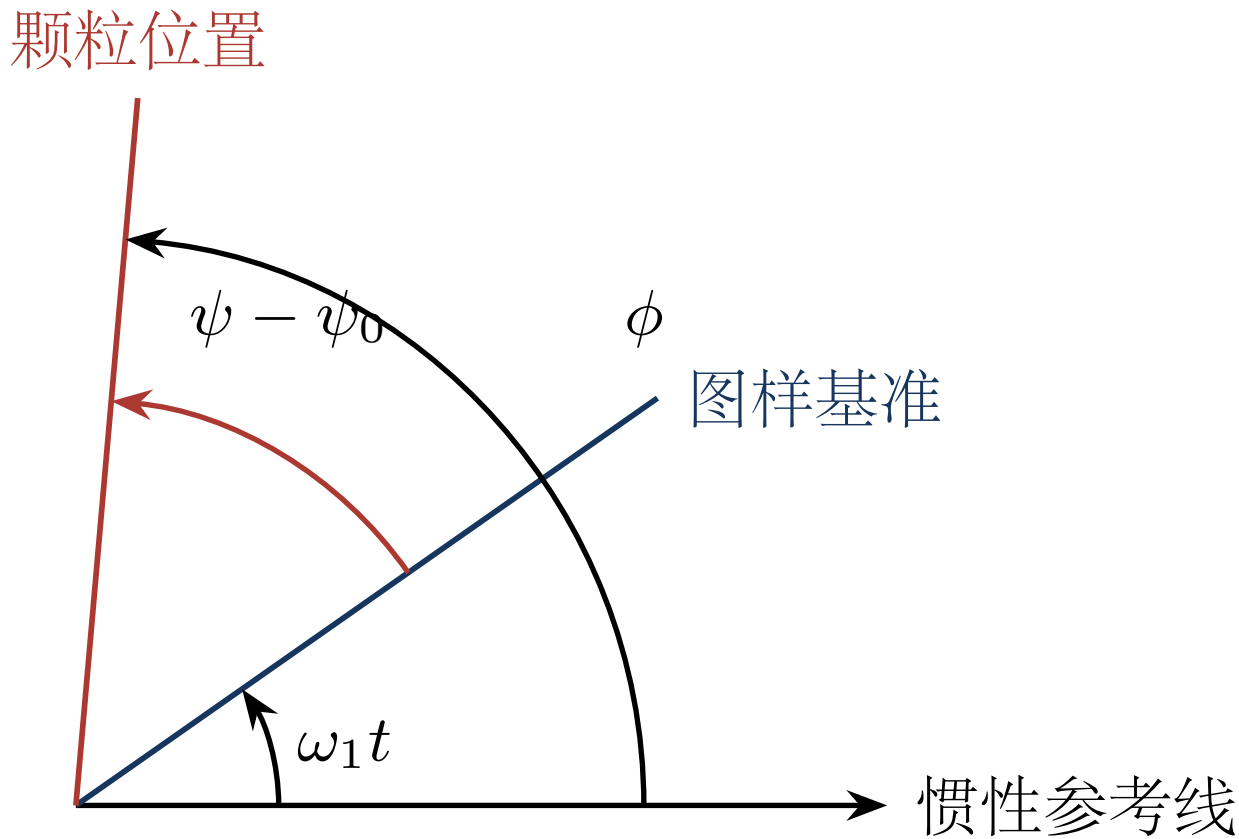

$$\psi = \phi - \omega_1 t + \psi_0$$

沿颗粒：$\dot{\psi} \simeq \omega_0 - \omega_1$